

Dean Francis Tolero

+971-56-138-4834 | deanfrancistolero@gmail.com | [linkedin.com/in/deanfrancistolero/](https://www.linkedin.com/in/deanfrancistolero/) | github.com/Dean-Francis-Tolero

EDUCATION

Canadian University Dubai

Dubai, United Arab Emirates

Bachelor in Computer Science (GPA: 3.81/4.0)

Aug. 2022 – May 2026

- Graduated Magna Cum Laude | Winner, Canadian University Coding Competition 2022 | Dean's List | Scholar

EXPERIENCE

Artificial Intelligence Research Assistant

Sep. 2025 – May 2026

Dr. Sahil Garg, Canadian University Dubai

Dubai, United Arab Emirates

- Contributed to the “Osmotic Intelligence” framework for cloud-edge workload distribution, running **Reinforcement Learning** experiments and integrating **Explainable AI** via Integrated Gradients.
- Conducted cost-performance analysis for sustainable **LLM training** supporting a **\$250,000 AWS research grant** proposal.
- Supported research, technical analysis, and content structuring for an **IEEE submission** and a **Springer Nature book** on Agentic AI.

Software Engineer

May 2025 – Sep. 2025

MoVlogs, Vibes Family Management

Dubai, United Arab Emirates

- Designed and deployed three responsive production websites, MoVlogs.com, MakeFym.com, and MummyMo.com, with full cross-device compatibility.
- Automated LinkedIn outreach with Python, scaling client acquisition by **300%** and reducing manual workload by **80%**.
- Built a Python service for international phone number parsing and metadata extraction, detecting country of origin at **98% accuracy**.

Data Analyst

Nov. 2025 – Dec. 2025

Namat Events Managing & Organizing

Dubai, United Arab Emirates

- Managed and analyzed real-time transportation data during live events, supporting logistics and dispatch decisions for the transportation team.
- Collaborated directly with **Patek Philippe**, ensuring client requirements and service standards were met.

PROJECTS

Flare, Federated Learning Phishing Defense | *Python, PyTorch, FastAPI, SQLAlchemy*

Sep. 2025 – May 2026

- **Technical Lead, 4-person team.** Architected a federated learning system where **raw email data never leaves the client device**, spanning a central aggregation server, a local inference and training server, and a Chrome MV3 extension.
- Fine-tuned DistilBERT, a 67M parameter transformer, for binary phishing classification, reaching **96.3% accuracy** and **0.963 F1** on a held-out test set from a 10K labeled email corpus.
- Built a FastAPI aggregation server implementing **FedAvg** over transmitted weight deltas, weighting client updates by local sample count, with a hybrid threshold/timeout round scheduler and SQLAlchemy-backed persistence.
- Built a local FastAPI inference server with **atomic model hot-reload** under a threading.Lock, on-device fine-tuning triggered by user flag volume, and an admin monitoring dashboard.
- Wrote a pytest suite covering FedAvg correctness, stratified split validation, and end-to-end API integration using in-memory SQLite fixtures and a stubbed detector.

Dementia MRI Classifier | *Python, PyTorch, NiBabel, Captum, scikit-learn*

Oct. 2025 – Dec. 2025

- Built a pipeline converting **OASIS 3D NIFTI** brain volumes into 2D axial slices, with Clinical Dementia Rating based labeling and quality filtering on intensity and contrast.
- Designed a lightweight **4-block CNN** of roughly **100K parameters** using batch normalization, 50% dropout, adaptive average pooling, and He initialization, trained with class-weighted CrossEntropyLoss on imbalanced data.
- Implemented **Integrated Gradients** attribution with **deletion and insertion metrics** over top-quartile pixels, surfacing hippocampal and ventricular regions driving predictions.
- Engineered a reproducible workflow with ReduceLROnPlateau scheduling, early stopping, gradient clipping, and fixed seeds for deterministic reruns.

RFID Attendance System | *JavaScript, Node.js, Express.js, MySQL, JWT, Arduino*

March 2025 – May 2025

- Developed and sold a production RFID attendance system to a school, generating approximately **1,000 AED** in revenue.
- Built a Node.js and Express REST backend with **JWT-authenticated role-based access**, serving separate teacher and parent portals against a normalized MySQL schema for students, scans, and attendance history.
- Integrated **solar-powered RFID scanner hardware** with the server, triggering **automated SMS notifications** to parents on scan events and generating attendance reports.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, C++, Java

ML and Data: PyTorch, HuggingFace Transformers, Captum, scikit-learn, NumPy, pandas, NiBabel

Backend and Systems: FastAPI, Node.js, Express, Flask, SQLAlchemy, REST APIs, Git, Docker, Linux, pytest

CERTIFICATIONS

Introduction to Modern AI | *Cisco*

Apr. 2026

Apply AI: Analyze Customer Reviews | *Cisco*

May 2026